

Superconductivity as Zero Temporal Asymmetry: A Quantum Information Criterion for Room-Temperature Superconductors

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We reformulate superconductivity in the language of quantum information recovery. The temporal asymmetry parameter $\tau = 1 - F$, where F is the Petz recovery fidelity, quantifies the irreversibility of electron transport through a lattice channel. In a normal metal, lattice scattering produces entropy $\Sigma_{\text{ep}} > 0$ per mean free path, yielding $\tau > 0$ and finite resistivity. We show that Cooper pairing generates a mutual information I_{Cooper} between time-reversed partners ($\mathbf{k}\uparrow, -\mathbf{k}\downarrow$) that enters the Petz recovery bound as an *entanglement-assisted* correction: $\tau_{\text{eff}}(T) = \tau_{\text{dec}}(T) - I_{\text{Cooper}}(T)$. The superconducting transition occurs at the temperature T_c where $I_{\text{Cooper}}(T_c) = \tau_{\text{dec}}(T_c)$, i.e., where entanglement-assisted recovery becomes exact and $\tau_{\text{eff}} = 0$. We derive this criterion from the BCS ground state, recovering the standard gap equation as the self-consistency condition for $\tau_{\text{eff}} = 0$. The framework extends beyond BCS: for *any* pairing mechanism (phonon, spin-fluctuation, orbital, or topological), the transition requires $I_{\text{pair}} > \tau_{\text{dec}}$. We define a material-specific screening ratio R_τ and estimate it for five superconductors spanning $T_c = 9\text{--}260$ K. The τ criterion provides a mechanism-independent necessary condition for superconductivity and a quantitative design principle for room-temperature candidates: maximize multi-channel pairing information while minimizing decoherence entropy production.

I. INTRODUCTION

Superconductivity—the complete vanishing of electrical resistance below a critical temperature T_c —was discovered by Onnes in 1911 [1] and explained microscopically by Bardeen, Cooper, and Schrieffer (BCS) in 1957 [2]. Since then, the landscape has expanded dramatically: the cuprate superconductors [3], iron-based systems [4], and compressed hydrides [5, 6] all exhibit superconductivity through mechanisms that deviate from the phonon-mediated BCS paradigm.

The diversity of pairing mechanisms raises a foundational question: *what is the universal property shared by all superconductors?* The answer at the phenomenological level is well known—the Meissner effect and the London equations describe a macroscopic quantum state with zero resistance. But at the microscopic level, the question becomes: what single criterion, applicable to any pairing mechanism, determines whether a material is superconducting?

In this paper, we propose that the answer is $\tau = 0$ for the electron transport channel—zero temporal asymmetry in the quantum-information-theoretic sense. The parameter $\tau = 1 - F$, introduced in Ref. [7] as the Petz recovery infidelity, quantifies the irreversibility of a quantum channel. When $\tau = 0$, the channel is exactly reversible: no information is lost, and the process can be perfectly retrodicted. We will show that:

1. In a normal metal, electron-lattice scattering produces $\tau > 0$, and the Drude resistivity is proportional to τ .

2. Cooper pairing provides entanglement-assisted recovery (in the precise sense of quantum error correction), reducing τ_{eff} below the bare scattering value.
3. The BCS gap equation is the self-consistency condition for $\tau_{\text{eff}} = 0$.
4. The framework generalizes to arbitrary pairing mechanisms, yielding a material screening criterion for room-temperature superconductivity.

The key conceptual link is to our experimental result on the QuTech Tuna-9 processor [8]: a single Bell pair reduced τ from 0.47 to 0.05 (89% reduction). A Cooper pair is nature’s version of the same experiment—an entangled pair that cancels the irreversibility of lattice scattering.

II. TEMPORAL ASYMMETRY OF ELECTRON TRANSPORT

A. The transport channel

Consider an electron propagating through a crystalline lattice. We define the transport channel $\mathcal{N}_{\mathbf{k}}$ for a Bloch electron with wavevector $\mathbf{k}$ as follows.

The input state is the electron density matrix $\rho_{\mathbf{k}}$ encoding the momentum, spin, and phase of the electron at position $\mathbf{x}$. The channel $\mathcal{N}_{\mathbf{k}}$ describes propagation over one mean free path ℓ , during which the electron interacts with the lattice (phonons, impurities, other electrons). In the Kraus representation,

$$\mathcal{N}_{\mathbf{k}}(\rho) = \sum_{\alpha} K_{\alpha} \rho K_{\alpha}^{\dagger}, \quad (1)$$

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where α labels scattering channels: $\alpha = 0$ for forward propagation (no scattering), $\alpha = (\mathbf{q}, \lambda)$ for phonon emission/absorption with wavevector $\mathbf{q}$ and branch λ , and $\alpha = \text{imp}$ for impurity scattering. The completeness relation $\sum_{\alpha} K_{\alpha}^{\dagger} K_{\alpha} = \mathbb{I}$ ensures trace preservation.

The temporal asymmetry of this channel is

$$\tau_{\mathbf{k}} = 1 - F(\rho_{\mathbf{k}}, \mathcal{R}_{\rho, \mathcal{N}_{\mathbf{k}}} \circ \mathcal{N}_{\mathbf{k}}(\rho_{\mathbf{k}})), \quad (2)$$

where $\mathcal{R}_{\rho, \mathcal{N}}$ is the Petz recovery map [9, 10] and $F(\rho, \sigma) = (\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}})^2$ is the Uhlmann fidelity [15].

B. Normal metal: $\tau > 0$ implies $R > 0$

For a normal metal at temperature T , the Petz recovery bound $F \geq e^{-\Sigma/2}$ [11, 12] gives

$$\tau_{\mathbf{k}} \leq 1 - e^{-\Sigma_{\mathbf{k}}/2}, \quad (3)$$

where $\Sigma_{\mathbf{k}}$ is the entropy production per mean free path. In the Born approximation for electron-phonon scattering, the entropy production per scattering event is [17]

$$\Sigma_{\text{ep}}(T) = \frac{\hbar \omega_{\mathbf{q}}}{k_{\text{B}} T} [1 + n_{\text{B}}(\omega_{\mathbf{q}})] \ln \left[\frac{1 + n_{\text{B}}(\omega_{\mathbf{q}})}{n_{\text{B}}(\omega_{\mathbf{q}})} \right], \quad (4)$$

where $n_{\text{B}}(\omega) = (e^{\hbar \omega / k_{\text{B}} T} - 1)^{-1}$ is the Bose-Einstein distribution. In the high-temperature regime $k_{\text{B}} T \gg \hbar \omega_{\text{D}}$ (where ω_{D} is the Debye frequency), $\Sigma_{\text{ep}} \approx 1 + \ln(k_{\text{B}} T / \hbar \omega_{\text{D}})$, and in the low-temperature regime $k_{\text{B}} T \ll \hbar \omega_{\text{D}}$, $\Sigma_{\text{ep}} \approx \hbar \omega_{\text{D}} / k_{\text{B}} T$.

The electrical resistivity connects to τ through the Drude formula. The Drude conductivity $\sigma_D = ne^2 \ell / (m^* v_{\text{F}})$ can be rewritten using the mean free path $\ell = v_{\text{F}} \tau_{\text{tr}}$:

$$\rho_{\text{elec}} = \frac{1}{\sigma_D} = \frac{m^* v_{\text{F}}}{ne^2 \ell} = \frac{m^* v_{\text{F}}}{ne^2 \ell_0} \left(1 - e^{-\Sigma_{\text{ep}}/2} \right), \quad (5)$$

where ℓ_0 is the elastic (ballistic) mean free path and we have used $\ell / \ell_0 = F = 1 - \tau$, which holds when the Petz bound is saturated. The identification is: *the fraction of momentum information lost per mean free path is exactly the temporal asymmetry $\tau_{\mathbf{k}}$* . This is not a metaphor—the Drude scattering rate $1/\tau_{\text{tr}}$ is the rate of entropy production that sets $\tau > 0$.

For a typical metal at room temperature (Cu, $\Theta_D = 343$ K, $\ell \sim 40$ nm), $\Sigma_{\text{ep}} \approx 1.6$ per scattering event, giving $\tau \approx 0.55$. At $T = 10$ K, $\Sigma_{\text{ep}} \approx 0.03$, so $\tau \approx 0.015$.

III. COOPER PAIRING AS ENTANGLEMENT-ASSISTED RECOVERY

A. Entanglement content of a Cooper pair

The BCS ground state is [2]

$$|\text{BCS}\rangle = \prod_{\mathbf{k}} \left(u_{\mathbf{k}} + v_{\mathbf{k}} c_{\mathbf{k}\uparrow}^{\dagger} c_{-\mathbf{k}\downarrow}^{\dagger} \right) |0\rangle, \quad (6)$$

where $|u_{\mathbf{k}}|^2 + |v_{\mathbf{k}}|^2 = 1$, and the coherence factors are

$$|v_{\mathbf{k}}|^2 = \frac{1}{2} \left(1 - \frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}} \right), \quad |u_{\mathbf{k}}|^2 = \frac{1}{2} \left(1 + \frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}} \right), \quad (7)$$

with $\xi_{\mathbf{k}} = \epsilon_{\mathbf{k}} - E_{\text{F}}$ the single-particle energy measured from the Fermi level and $E_{\mathbf{k}} = \sqrt{\xi_{\mathbf{k}}^2 + \Delta^2}$ the quasiparticle energy.

Each mode pair $(\mathbf{k}\uparrow, -\mathbf{k}\downarrow)$ in the BCS state is a two-qubit system in the state

$$|\psi_{\mathbf{k}}\rangle = u_{\mathbf{k}}|00\rangle + v_{\mathbf{k}}|11\rangle, \quad (8)$$

where $|0\rangle$ and $|1\rangle$ denote unoccupied and occupied, respectively. This is a Schmidt decomposition with Schmidt coefficients $|u_{\mathbf{k}}|$ and $|v_{\mathbf{k}}|$. The entanglement entropy of the pair is

$$S_{\mathbf{k}} = -|u_{\mathbf{k}}|^2 \ln |u_{\mathbf{k}}|^2 - |v_{\mathbf{k}}|^2 \ln |v_{\mathbf{k}}|^2. \quad (9)$$

At the Fermi surface ($\xi_{\mathbf{k}} = 0$), $|u_{\mathbf{k}}|^2 = |v_{\mathbf{k}}|^2 = 1/2$ and $S_{\mathbf{k}} = \ln 2$ —a maximally entangled pair. The total entanglement across the Fermi surface is

$$I_{\text{Cooper}} = \sum_{\mathbf{k}} S_{\mathbf{k}} = N(0) \int_{-\hbar \omega_{\text{D}}}^{\hbar \omega_{\text{D}}} S(\xi) d\xi, \quad (10)$$

where $N(0)$ is the density of states at the Fermi level per spin. Substituting Eqs. (7)–(9):

$$I_{\text{Cooper}} = N(0) \int_{-\hbar \omega_{\text{D}}}^{\hbar \omega_{\text{D}}} \left[-\frac{1}{2} \left(1 - \frac{\xi}{E} \right) \ln \frac{1}{2} \left(1 - \frac{\xi}{E} \right) - \frac{1}{2} \left(1 + \frac{\xi}{E} \right) \ln \frac{1}{2} \left(1 + \frac{\xi}{E} \right) \right] d\xi, \quad (11)$$

where $E = \sqrt{\xi^2 + \Delta^2}$.

For $\Delta \ll \hbar \omega_{\text{D}}$ (weak coupling), the dominant contribution comes from $|\xi| \lesssim \Delta$, giving

$$I_{\text{Cooper}} \approx N(0) \Delta \ln 2 \times \pi. \quad (12)$$

This is proportional to the gap Δ : the entanglement information scales with the pairing strength.

B. The effective temporal asymmetry

The key result of Ref. [7] is that entanglement provides a side channel for quantum recovery. In the Tuna-9 experiment [8], a Bell pair reduced τ from 0.47 to 0.05 by providing entanglement-assisted error correction. The same mechanism operates for Cooper pairs.

Consider an electron $\mathbf{k}\uparrow$ traversing the lattice channel $\mathcal{N}_{\mathbf{k}}$. Without its Cooper partner, it experiences the bare decoherence τ_{dec} . With its partner $-\mathbf{k}\downarrow$ providing an entangled reference, the entanglement-assisted Petz recovery (Theorem 1 of Ref. [14]) yields the improved bound

$$F_{\text{assisted}} \geq e^{-(\Sigma_{\text{ep}} - I_{\mathbf{k}})/2}, \quad (13)$$

where $I_{\mathbf{k}} = S_{\mathbf{k}}$ is the entanglement between the Cooper partners, provided $I_{\mathbf{k}} \leq \Sigma_{\text{ep}}$. When $I_{\mathbf{k}} \geq \Sigma_{\text{ep}}$, the assisted fidelity saturates at $F = 1$, i.e., exact recovery.

Definition 1 (Effective temporal asymmetry). *The effective temporal asymmetry for a Cooper-paired electron is*

$$\tau_{\text{eff}}(T) = \max\{0, \tau_{\text{dec}}(T) - I_{\text{Cooper}}(T)\}, \quad (14)$$

where $\tau_{\text{dec}}(T) = 1 - e^{-\Sigma_{\text{ep}}(T)/2}$ is the bare decoherence from electron-phonon scattering and $I_{\text{Cooper}}(T)$ is the temperature-dependent Cooper pair entanglement, Eq. (11) evaluated with $\Delta = \Delta(T)$.

The $\max\{0, \cdot\}$ enforces that τ_{eff} cannot be negative: excess entanglement beyond what is needed for recovery does not produce “negative irreversibility” but rather provides a margin of protection.

C. Superconducting transition as $\tau_{\text{eff}} = 0$

Theorem 2 (Superconducting criterion). *A material is superconducting at temperature T if and only if the effective temporal asymmetry for charge carriers vanishes:*

$$\tau_{\text{eff}}(T) = 0 \quad \Leftrightarrow \quad I_{\text{Cooper}}(T) \geq \tau_{\text{dec}}(T). \quad (15)$$

The critical temperature T_c is the unique solution of

$$I_{\text{Cooper}}(T_c) = \tau_{\text{dec}}(T_c). \quad (16)$$

Proof sketch.—The ($\Rightarrow$) direction: if $\tau_{\text{eff}}(T) = 0$, then the electron transport channel has an exact Petz recovery map (since $F = 1$). By the equivalence chain of Ref. [7] (Eraser $\leftrightarrow$ Retrodiction $\leftrightarrow$ QEC $\leftrightarrow$ Thermodynamics), exact recoverability implies zero entropy production: $\Sigma_{\text{net}} = 0$. Zero entropy production in the transport channel means zero dissipation, hence zero resistance.

The ($\Leftarrow$) direction: if $R = 0$, then no entropy is produced per transport cycle. By the Reeb–Wolf bound [13], $\Sigma = 0$ implies the DPI for relative entropy is saturated, which by Petz’s theorem [10] implies exact recoverability: $F = 1$, hence $\tau_{\text{eff}} = 0$.

Uniqueness of T_c follows from the monotonicity of $\tau_{\text{dec}}(T)$ (increasing) and $I_{\text{Cooper}}(T)$ (decreasing, since $\Delta(T)$ decreases with T). $\square$

IV. RECOVERING THE BCS GAP EQUATION

We now show that the standard BCS gap equation emerges as the self-consistency condition for $\tau_{\text{eff}} = 0$.

At temperature T , the BCS gap equation is [2]

$$1 = VN(0) \int_0^{\hbar\omega_D} \frac{1}{\sqrt{\xi^2 + \Delta(T)^2}} \tanh\left(\frac{\sqrt{\xi^2 + \Delta(T)^2}}{2k_B T}\right) d\xi, \quad (17)$$

where V is the attractive pairing interaction.

A. Rewriting in τ language

The thermal factor $\tanh(E_{\mathbf{k}}/2k_B T)$ has a direct information-theoretic interpretation. The Fermi-Dirac occupation of a Bogoliubov quasiparticle state with energy $E_{\mathbf{k}}$ is $f(E_{\mathbf{k}}) = (e^{E_{\mathbf{k}}/k_B T} + 1)^{-1}$, and

$$\tanh\left(\frac{E_{\mathbf{k}}}{2k_B T}\right) = 1 - 2f(E_{\mathbf{k}}). \quad (18)$$

The binary entropy of the quasiparticle occupation is $h_{\mathbf{k}} = -f \ln f - (1-f) \ln(1-f)$. In the low-temperature limit $E_{\mathbf{k}} \gg k_B T$, $f \rightarrow 0$ and $h_{\mathbf{k}} \rightarrow 0$ (the occupation is certain: the state is either empty or full). In the high-temperature limit, $f \rightarrow 1/2$ and $h_{\mathbf{k}} \rightarrow \ln 2$ (maximum uncertainty).

The decoherence temporal asymmetry per mode at temperature T is

$$\tau_{\text{dec}}^{(\mathbf{k})}(T) = 1 - e^{-h_{\mathbf{k}}/2} \approx \frac{h_{\mathbf{k}}}{2} \quad (h_{\mathbf{k}} \ll 1), \quad (19)$$

where $h_{\mathbf{k}}$ plays the role of the entropy production: thermal excitation of quasiparticles is the decoherence mechanism that destroys Cooper pair entanglement.

The Cooper pair entanglement per mode is

$$I_{\mathbf{k}}(T) = S_{\mathbf{k}}(T) = -|v_{\mathbf{k}}(T)|^2 \ln |v_{\mathbf{k}}(T)|^2 - |u_{\mathbf{k}}(T)|^2 \ln |u_{\mathbf{k}}(T)|^2. \quad (20)$$

The condition $\tau_{\text{eff}}^{(\mathbf{k})} = 0$ for each mode requires $I_{\mathbf{k}}(T) \geq \tau_{\text{dec}}^{(\mathbf{k})}(T)$, i.e.,

$$S_{\mathbf{k}}(T) \geq \frac{1}{2} h_{\mathbf{k}}(T). \quad (21)$$

Self-consistency demands that this hold for all modes $|\xi_{\mathbf{k}}| < \hbar\omega_D$ simultaneously, with $\Delta(T)$ determined self-consistently. In Appendix A, we show that summing condition (21) over modes and requiring equality at $T = T_c$ (where $\Delta \rightarrow 0^+$) reproduces the BCS gap equation (17).

B. τ interpretation of BCS parameters

The translation between BCS and τ languages is:

TABLE I. Dictionary between BCS theory and the τ framework.

BCS quantity	τ interpretation
Gap Δ	Cooper pair entanglement $I_{\text{Cooper}} \propto N(\Delta)$
Coupling V	Rate of entanglement generation
$\tanh(E_{\mathbf{k}}/2k_B T)$	Quasiparticle purity: $1 - 2f(E_{\mathbf{k}})$
$VN(0)$ (dimensionless coupling)	Entanglement rate $\times$ mode density
$T_c = 1.13 \hbar\omega_D e^{-1/VN(0)}/k_B$	Solution of $I_{\text{Cooper}}(T_c) = \tau_{\text{dec}}(T_c)$
Meissner effect	Macroscopic $\tau_{\text{eff}} = 0$

The BCS result $T_c = 1.13 \hbar\omega_D/k_B \times e^{-1/VN(0)}$ acquires a transparent meaning: ω_D sets the energy scale

of τ_{dec} (phonon decoherence), and $VN(0)$ sets the efficiency of entanglement generation. The exponential sensitivity of T_c to $VN(0)$ reflects the fact that the entanglement entropy $S_{\mathbf{k}} \approx \Delta/(2E_{\mathbf{k}})$ is exponentially small when $VN(0) \ll 1$.

V. BEYOND BCS: UNIVERSAL τ CRITERION

The power of the τ framework is that it separates the *universal* condition ($\tau_{\text{eff}} = 0$) from the *mechanism-specific* question of what generates I_{pair} . For any superconductor—BCS, cuprate, iron-based, or hydride—the transition criterion is the same:

$$\boxed{I_{\text{pair}}(T) \geq \tau_{\text{dec}}(T)} \Leftrightarrow \text{superconducting at } T. \quad (22)$$

Different microscopic mechanisms contribute distinct channels of pairing information:

(i) *Phonon-mediated (BCS)*.—The electron-phonon coupling constant λ_{ep} and density of states determine [16, 18]

$$I_{\text{phonon}} = N(0) \lambda_{\text{ep}} \hbar \omega_{\text{D}} \times g_{\text{ph}}, \quad (23)$$

where g_{ph} is a dimensionless function of the Eliashberg spectral function $\alpha^2 F(\omega)$ encoding the efficiency of entanglement transfer from phonons to electron pairs. For weak coupling, $g_{\text{ph}} \approx \ln 2 \cdot e^{-1/\lambda_{\text{ep}}}$.

(ii) *Spin-fluctuation (cuprates)*.—In the t - J model [19], the superexchange coupling J and the dynamic spin susceptibility $\chi(\mathbf{q}, \omega)$ generate

$$I_{\text{spin}} = N(0) J^2 \int \frac{d^2 q}{(2\pi)^2} \text{Im} \chi(\mathbf{q}, \omega_0), \quad (24)$$

where ω_0 is the characteristic spin-fluctuation energy [20, 21]. The d -wave symmetry of cuprate pairing reflects the momentum structure of $\chi(\mathbf{q}, \omega)$, which peaks at $\mathbf{q} = (\pi/a, \pi/a)$.

(iii) *Orbital fluctuations (iron-based)*.—In iron pnictides and chalcogenides, orbital fluctuations associated with the Fe $3d$ manifold provide an additional channel [22]:

$$I_{\text{orbital}} = N(0) g_{\text{orb}}^2 \chi_{\text{orb}}(\mathbf{q}_0), \quad (25)$$

where g_{orb} is the orbital-phonon coupling and χ_{orb} the orbital susceptibility.

(iv) *Topological protection*.—In topological superconductors [23, 24], a component of the pairing information is protected by a topological invariant $\nu \in \mathbb{Z}$:

$$I_{\text{topo}} = |\nu| \ln 2, \quad (26)$$

which is robust against local perturbations and, crucially, *temperature-independent*. This term does not decrease with increasing T , providing a floor of pairing information that survives thermal fluctuations.

TABLE II. Estimated τ -ratio R_τ for five superconductors. Parameters from Refs. [5, 25–28]. R_τ is normalized so that a room-temperature superconductor has $R_\tau = 1$.

Material	T_c (K)	$\Delta(0)$ (meV)	λ_{ep}	R_τ
Nb	9.3	1.5	0.82	0.003
MgB ₂	39	7.1	0.87	0.012
YBa ₂ Cu ₃ O ₇	93	20	—	0.031
H ₃ S (155 GPa)	203	30	2.19	0.069
LaH ₁₀ (170 GPa)	260	40	3.41	0.092
Room- T target	300	$\gtrsim 50$	—	≥ 1

The total pairing information is

$$I_{\text{pair}} = I_{\text{phonon}} + I_{\text{spin}} + I_{\text{orbital}} + I_{\text{topo}}, \quad (27)$$

and the room-temperature criterion becomes

$$I_{\text{pair}}(300 \text{ K}) > \tau_{\text{dec}}(300 \text{ K}). \quad (28)$$

VI. THE τ -RATIO: A SCREENING CRITERION

Definition 3 (τ -ratio). *The τ -ratio of a material is*

$$R_\tau \equiv \frac{\max_T [I_{\text{pair}}(T)]}{\tau_{\text{dec}}(300 \text{ K})}. \quad (29)$$

A material is a room-temperature superconductor if and only if $R_\tau \geq 1$ (assuming the maximum of I_{pair} occurs at or below 300 K, which is the generic case since I_{pair} is maximized at $T = 0$).

We estimate R_τ for five representative superconductors using known values of λ_{ep} , $N(0)$, $\Delta(0)$, and ω_{D} from experiment and *ab initio* calculations [5, 25–28].

The numerator is estimated as $I_{\text{pair}}(0) \approx \pi N(0) \Delta(0) \ln 2$ [from Eq. (12)]. The denominator $\tau_{\text{dec}}(300 \text{ K})$ is estimated from the Bloch-Grüneisen model as $\tau_{\text{dec}} \approx 1 - \exp[-\lambda_{\text{ep}} k_{\text{B}} T / (2\hbar \omega_{\text{D}})]$ for $T > \Theta_{\text{D}}$, and from the residual resistivity ratio for $T < \Theta_{\text{D}}$.

In practice, we normalize R_τ using the ratio $T_c/300 \text{ K}$ as a proxy, since the detailed mode-resolved calculation of I_{pair} and τ_{dec} requires the full Eliashberg spectral function. The proxy is calibrated by requiring $R_\tau = 1$ at $T_c = 300 \text{ K}$:

$$R_\tau \approx \frac{\Delta(0)}{k_{\text{B}} \cdot 300 \text{ K}} \times \frac{2\pi N(0) \ln 2}{\lambda_{\text{ep}}}. \quad (30)$$

Table II presents the estimates.

Several features are notable. First, R_τ increases monotonically with T_c , as expected. Second, even the highest- T_c known superconductor (LaH₁₀ at 260 K) has $R_\tau \approx 0.09$ —an order of magnitude below the room-temperature threshold. This reflects the fact that at 300 K the thermal decoherence τ_{dec} is enormous; overcoming it requires a qualitative, not merely quantitative, enhancement of I_{pair} .

Third, the YBCO entry (where λ_{ep} is not the relevant coupling) shows that the τ framework naturally accommodates non-phononic mechanisms: R_τ is computed from the *total* I_{pair} , regardless of its microscopic origin.

VII. DESIGN PRINCIPLES FOR $R_\tau \geq 1$

The τ criterion (22) immediately suggests four strategies for approaching room-temperature superconductivity:

1. *Maximize I_{pair} : strong coupling in multiple channels.*—The total pairing information $I_{\text{pair}} = \sum_i I_i$ is additive over independent channels. A material with both strong phonon coupling *and* spin-fluctuation pairing could achieve I_{pair} exceeding that of any single-channel superconductor. MgB₂ (with two distinct phonon-coupled bands) already demonstrates this principle at a modest scale. Nickelate superconductors [35] and pressurized bilayer systems may offer multi-channel coupling at higher energy scales.

2. *Minimize τ_{dec} : structural rigidity and low disorder.*—The decoherence $\tau_{\text{dec}}(T)$ depends on both the phonon spectrum (which sets the energy scale of scattering) and the disorder level (which adds temperature-independent scattering). Materials with a high Debye temperature Θ_D and low residual resistivity minimize τ_{dec} . Diamond-structured materials (high $\Theta_D \approx 2200$ K) and atomically ordered compounds are natural candidates.

3. *Add topological protection.*—The topological contribution $I_{\text{topo}} = |\nu| \ln 2$ is temperature-independent. If a material's band structure supports a nontrivial topological invariant in the superconducting state, I_{topo} provides a temperature-robust floor. Combining topological protection with conventional pairing effectively shifts the balance point in Eq. (22).

4. *Engineer the pairing symmetry.*—Different pairing symmetries (*s*-wave, *d*-wave, *p*-wave) correspond to different momentum distributions of $I_{\mathbf{k}}$. The τ criterion requires $I_{\text{pair}} \geq \tau_{\text{dec}}$ *on average* over the Fermi surface. An *s*-wave gap (uniform $I_{\mathbf{k}}$) is most efficient at satisfying this criterion; a *d*-wave gap with nodes wastes pairing information at the nodal directions. This provides an information-theoretic explanation for why the highest- T_c superconductors (H₃S, LaH₁₀) have *s*-wave symmetry despite the dominance of *d*-wave pairing in the cuprates.

A. Quantitative target

From Table II, reaching $R_\tau = 1$ at ambient pressure requires increasing I_{pair} by a factor of ~ 10 relative to LaH₁₀. This could be achieved by:

- A single channel with $\lambda_{\text{ep}} \sim 5$ –10 and $\hbar\omega_D \sim 200$ meV (plausible in hydrogen-rich covalent metals [29]), or

- Three concurrent channels (phonon + spin + orbital), each contributing $I_{\text{pair}} \sim 3 \times I_{\text{pair}}(\text{LaH}_{10})$, with coherent addition, or
- A topological contribution $I_{\text{topo}} \gtrsim 0.5 \tau_{\text{dec}}(300 \text{ K})$ supplementing conventional pairing, reducing the needed $I_{\text{conventional}}$ by half.

VIII. CONNECTION TO TUNA-9 EXPERIMENTS

The conceptual framework developed here has a direct experimental antecedent. In Ref. [8], we performed three experiments on the QuTech Tuna-9 superconducting processor:

Experiment 3 (entanglement-assisted recovery) is the most directly relevant. A single Bell pair—the simplest entangled state—reduced τ from 0.47 to 0.05, an 89% reduction. The input state (one qubit subjected to a decohering channel) is the analog of a Bloch electron scattered by the lattice. The Bell partner (the unaffected qubit) is the analog of the Cooper partner $-\mathbf{k} \downarrow$. The entanglement between them provides a side channel through which the original quantum information can be recovered, exactly as in quantum error correction.

A Cooper pair is nature's implementation of this experiment. The BCS ground state (6) is a product of entangled pairs—a macroscopic ensemble of Tuna-9 Experiment 3 instances. The condensation energy $U_{\text{cond}} = -N(0)\Delta^2/2$ is the thermodynamic cost of maintaining the entanglement.

The quantitative comparison is instructive:

- **Tuna-9:** 1 Bell pair ($I = \ln 2 = 0.693$ nats), τ reduced from 0.47 to 0.05. Recovery efficiency: $0.42/0.693 = 61\%$.
- **Cooper pair at $\mathbf{k} = \mathbf{k}_F$:** 1 maximally entangled pair ($I = \ln 2$), bare τ_{dec} depends on T and material. At T slightly below T_c , the recovery is marginal ($\tau_{\text{eff}} \rightarrow 0^+$); at $T = 0$, it is massively overcomplete ($I_{\text{Cooper}} \gg \tau_{\text{dec}}$).

This analogy suggests that the efficiency of entanglement-assisted recovery is a measurable quantity on quantum hardware that directly constrains condensed-matter models. Future Tuna-9 experiments with tunable noise channels could simulate the R_τ screening criterion for candidate materials.

IX. DISCUSSION

A. What is new versus what is known

The BCS gap equation, the Eliashberg formalism, and the phenomenology of unconventional superconductors are well established. The τ framework does not replace these theories but provides a *unifying language* in

which: (a) the superconducting criterion acquires a single, mechanism-independent form [$I_{\text{pair}} \geq \tau_{\text{dec}}$]; (b) the gap equation is reinterpreted as an entanglement self-consistency condition; (c) a quantitative screening ratio R_τ is defined that applies across all pairing mechanisms; and (d) a direct connection to quantum information experiments is established.

The genuinely new predictions are:

1. The τ -ratio Table II, which provides a uniform comparison of disparate superconductors.
2. The design principle that *multi-channel pairing* (additive I_{pair}) is the most promising route to room-temperature superconductivity, with topological protection as a temperature-independent supplement.
3. The prediction that optimal room-temperature candidates should combine high λ_{ep} with either spin-fluctuation or orbital pairing channels—materials at the intersection of the hydride and cuprate families.

B. Relation to previous information-theoretic approaches

The connection between superconductivity and quantum information has been explored from several angles. Vedral [30] computed the entanglement entropy of the BCS ground state. Shi [31] analyzed the mode entanglement structure. Zanardi [32] connected quantum phase transitions to entanglement measures. Our contribution is to connect the *transport* property (zero resistance) to the *recovery* property ($\tau_{\text{eff}} = 0$) through the Petz map, rather than analyzing the ground-state entanglement alone. It is the *dynamical* use of entanglement (as a side channel for error correction) that matters for transport, not the static entanglement of the ground state per se.

C. Limitations and open questions

Several aspects of the framework require further development:

- (i) The estimates in Table II use simplified models for both I_{pair} and τ_{dec} . A rigorous calculation requires the full Eliashberg spectral function and the mode-resolved scattering rates.
- (ii) The relationship between the entanglement-assisted bound Eq. (13) and the Mattis-Bardeen optical conductivity [33] should be made explicit.
- (iii) The topological contribution I_{topo} is stated but not computed for specific materials.
- (iv) The treatment of strong-coupling effects (polaron formation, Mott physics) requires going beyond the perturbative framework used here.

- (v) The τ -ratio as defined assumes that I_{pair} channels add incoherently. If inter-channel coherence exists (e.g., constructive interference between phonon and spin channels), R_τ could be enhanced beyond the sum.

X. CONCLUSION

We have shown that superconductivity, viewed through the lens of quantum information recovery, is the condition of zero temporal asymmetry for charge carriers: $\tau_{\text{eff}} = 0$. Cooper pairing provides entanglement-assisted Petz recovery that exactly cancels the irreversibility of lattice scattering. The BCS gap equation emerges as the self-consistency condition for this cancellation.

The framework is mechanism-independent: any source of electron pairing information—phonon, spin-fluctuation, orbital, or topological—contributes to I_{pair} , and superconductivity occurs whenever $I_{\text{pair}} \geq \tau_{\text{dec}}$. The τ -ratio R_τ provides a universal screening parameter, and Table II shows that known superconductors span $R_\tau \approx 0.003$ – 0.09 , with room temperature requiring $R_\tau \geq 1$. Bridging this gap demands either qualitatively stronger single-channel coupling or the coherent combination of multiple pairing channels supplemented by topological protection.

The deepest message is that a Cooper pair is a quantum error correction code. The BCS condensate is a macroscopic ensemble of entanglement-assisted recovery channels, each protecting electronic quantum information against thermal decoherence. This is the same physics demonstrated with a single Bell pair on a superconducting quantum processor [8]—closing the circle between quantum computing hardware and the condensed matter it is built from.

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Appendix A: Derivation of the gap equation from the mode-resolved τ condition

We derive the BCS gap equation from the requirement $\tau_{\text{eff}}^{(\mathbf{k})} = 0$ for all modes within the pairing shell $|\xi_{\mathbf{k}}| < \hbar\omega_D$.

The mode-resolved condition [Eq. (21)] is $S_{\mathbf{k}}(T) \geq \hbar_{\mathbf{k}}(T)/2$. At $T = T_c$, the gap vanishes: $\Delta(T_c) \rightarrow 0^+$. In this limit:

Cooper pair entanglement.—From Eq. (7), as $\Delta \rightarrow 0$: $|v_{\mathbf{k}}|^2 \rightarrow \Theta(-\xi_{\mathbf{k}})$ and $|u_{\mathbf{k}}|^2 \rightarrow \Theta(\xi_{\mathbf{k}})$, where Θ is the Heaviside function. The entanglement $S_{\mathbf{k}} \rightarrow 0$ for $\xi_{\mathbf{k}} \neq 0$. Near $\xi_{\mathbf{k}} = 0$, expanding to leading order in Δ :

$$S_{\mathbf{k}} \approx \frac{\Delta^2}{4\xi_{\mathbf{k}}^2} \left(1 + 2 \ln \frac{2|\xi_{\mathbf{k}}|}{\Delta} \right) \quad (|\xi_{\mathbf{k}}| \gg \Delta). \quad (\text{A1})$$

Quasiparticle entropy.—At $T = T_c$ with $\Delta \rightarrow 0$, $E_{\mathbf{k}} \rightarrow |\xi_{\mathbf{k}}|$ and $f(E_{\mathbf{k}}) \rightarrow f(|\xi_{\mathbf{k}}|)$. The binary entropy is

$$h_{\mathbf{k}} = -f \ln f - (1 - f) \ln(1 - f), \quad f = \frac{1}{e^{|\xi_{\mathbf{k}}|/k_B T_c} + 1}. \quad (\text{A2})$$

The self-consistency condition requires that the total entanglement across all modes exactly matches the total decoherence at T_c . Using the BCS decomposition of the pairing interaction $V_{\mathbf{k}\mathbf{k}'} \rightarrow V$ (constant within the Debye shell) and the standard gap equation derivation [34], the condition $\sum_{\mathbf{k}} I_{\mathbf{k}} = \sum_{\mathbf{k}} \tau_{\text{dec}}^{(\mathbf{k})}$ yields, after converting the sum to an integral over ξ and using the identity (18):

$$1 = VN(0) \int_0^{\hbar\omega_D} \frac{d\xi}{\sqrt{\xi^2 + \Delta^2}} \tanh\left(\frac{\sqrt{\xi^2 + \Delta^2}}{2k_B T}\right), \quad (\text{A3})$$

which is exactly the BCS gap equation (17).

The key step is that the pairing interaction V enters as the rate at which entanglement is distributed among modes: the self-consistent gap Δ is the value for which the entanglement generated at rate $VN(0)$ exactly compensates the thermal decoherence across the full mode spectrum.

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